

Hamiltonian-Inspired Energy Dissipation and Chunk-Wise Variational State Coupling for Efficient Multimodal State-Space Models

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Abstract—State-Space Models (SSMs) such as Mamba and State-Space Duality (SSD) have emerged as computationally efficient, linear-time sequence modeling alternatives to quadratic multi-head self-attention. However, deploying state-space recurrence in multimodal vision-language retrieval introduces two fundamental challenges: (1) sensitivity to high-frequency representation noise and background token perturbations across long sequences, and (2) cross-modal representation redundancy where unaligned modality-specific boundary states degrade downstream gallery retrieval. In this work, we propose a principled multimodal state-space framework comprising two foundational components: (i) the Hamiltonian-Inspired Energy Dissipation Operator (HEDO), a discrete coordinate-momentum update with positive damping that introduces a dissipative inductive bias to suppress high-frequency perturbation noise, and (ii) Chunk-Wise Variational State Coupling (HVSC), which aggregates valid chunk boundary states via attention-mask weighting and regularizes modality-specific Gaussian latent spaces through symmetric Kullback-Leibler (KL) divergence in training while preserving strictly unimodal deterministic inference at test time. We evaluate the proposed architecture on the official Flickr8k benchmark (6,000 train, 1,000 validation, 1,000 held-out test images) using frozen ViT-B/16 and RoBERTa-base backbones across 12 controlled multi-seed runs (4 models $\times$ 3 seeds : 42, 43, 44). The full HEDO-HVSC model achieves a test Mean Recall of $48.28\% \pm 0.59\%$ ($26.73\% \pm 1.87\%$ I2T R@1, $20.71\% \pm 0.18\%$ T2I R@1) with guaranteed $O(L)$ linear scaling (2.89 ms at $L = 16$ to 38.60 ms at $L = 256$) and demonstrates enhanced robustness under severe pixel-space Gaussian noise ($\sigma = 0.10$).

Index Terms—Multimodal Retrieval, State Space Models, State Space Duality, Hamiltonian Dynamics, Variational State Coupling, Vision-Language Alignment.

I. INTRODUCTION

CROSS-MODAL image-text retrieval represents a foundational paradigm in vision-language understanding, serving critical applications including semantic search, visual question answering, and multimodal knowledge retrieval [1]–[3]. The primary challenge involves mapping heterogeneous visual patches and textual tokens into a shared metric embedding space where semantic similarity corresponds directly to cosine proximity [4], [5].

Predominant architectures have historically relied on Transformer-based cross-attention mechanisms [6]–[8]. While expressive, pairwise self-attention incurs quadratic computational complexity $O(L^2)$ and memory footprint with respect to sequence length L , severely restricting efficiency when scaling

Controlled 4-Model Factorial Benchmark on Flickr8k Test Set (N=3 Sees)

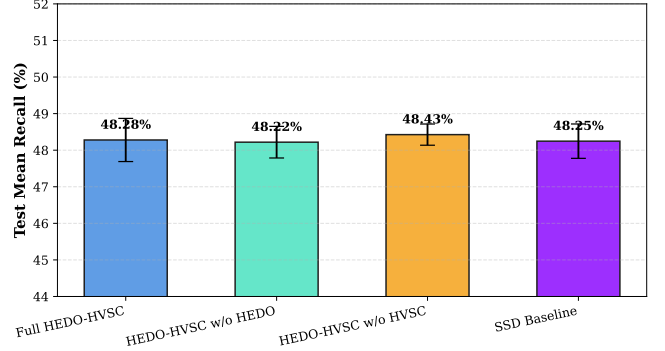


Fig. 1: Controlled 4-model factorial benchmark comparison across 3 random seeds (42, 43, 44) on the official Flickr8k held-out test split ($N = 1,000$ images, 5,000 captions). Error bars denote ± 1 standard deviation.

to fine-grained image patches and extended textual descriptions [9].

To circumvent quadratic scaling, State-Space Models (SSMs)—notably S4 [10], Mamba [11], and State-Space Duality (SSD) architectures [12]—have introduced continuous-time linear dynamical systems parameterized via input-dependent discrete recurrence. SSMs achieve strict $O(L)$ computation during sequential processing and enable linear-time recurrent inference.

Despite their theoretical appeal, applying state-space sequence blocks to multimodal representation alignment introduces two open technical hurdles:

- 1) **High-Frequency Representation Instability:** Discrete recurrent sequence transitions tend to propagate and amplify high-frequency token perturbations across sequence steps, making standard state recurrence vulnerable to visual clutter, brightness shifts, and pixel sensor noise [13], [14].
- 2) **Cross-Modal Boundary State Misalignment:** Independent unidirectional state-space recurrence across vision and language streams accumulates modality-specific drift. Padded tokens in textual sequences can severely corrupt aggregated boundary states if not correctly masked [15].

To address these challenges, we introduce **HEDO-HVSC**, a unified multimodal architecture formulated on two core mathematical principles:

1) **Hamiltonian-Inspired Energy Dissipation Operator (HEDO):**

A coordinate-momentum dynamical transformation parameterized by learned vector fields W_q, W_p , coupled with a fixed positive damping coefficient $\beta = 0.05$ and perturbation step $\gamma = 0.1$. HEDO introduces an explicit dissipative inductive bias that suppresses high-frequency representation noise prior to state-space recurrence.

2) **Chunk-Wise Variational State Coupling (HVSC):**

A state-continuous chunk-wise SSD block that partitions sequences into chunks of size $C = 16$, propagates recurrent hidden states h_k continuously across chunk boundaries ($h_{\text{start},k} = h_{\text{end},k-1}$), and aggregates boundary states via attention-mask weighting ($\mathbf{h}_{\text{bound}} = \frac{\sum m_k \mathbf{b}_k}{\sum m_k}$). Modality-specific Gaussian latent spaces are coupled through symmetric Kullback-Leibler (KL) divergence in FP32 precision during training, while test-time retrieval uses deterministic posterior means strictly independently per modality.

To ensure publication-level reproducibility, we evaluate HEDO-HVSC under an exhaustive 12-run factorial matrix (4 configurations $\times$ 3 seeds) on the official Flickr8k benchmark (6,000 train, 1,000 validation, 1,000 test images) with frozen ViT-B/16 and RoBERTa-base backbones.

II. RELATED WORK

A. Vision-Language Representation Learning

Cross-modal metric learning models such as CLIP [1] and ALIGN [3] project visual and linguistic features into a shared normalized embedding space using symmetric InfoNCE loss. However, these frameworks typically aggregate sequence features via simple pooling or Transformer self-attention, incurring significant computational overhead on longer visual sequences [2].

B. State Space Models for Sequential Modeling

Structured State Space Sequence models (S4) [10] parameterized linear time-invariant systems via HiPPO matrix initializations [16]. Mamba [11] introduced input-dependent selective state spaces, and Mamba-2 / SSD [12] established the theoretical duality between structured masked attention and continuous recurrence. In this work, we build upon chunk-wise recurrent state-space formulations to provide exact boundary state tracking for multimodal sequence coupling.

C. Energy-Based & Hamiltonian Neural Formulations

Hamiltonian Neural Networks (HNNs) [17] and Port-Hamiltonian Neural ODEs [18] model conservative physical dynamical systems via symplectic gradients:

$$\dot{q} = \frac{\partial \mathcal{H}}{\partial p}, \quad \dot{p} = -\frac{\partial \mathcal{H}}{\partial q} - \mathbf{D}p \quad (1)$$

Rather than claiming formal continuous-time symplectic conservation, our work draws inspiration from discrete dissipative mechanics to introduce a parameter-efficient token-level stabilizing operator that limits energy amplification.

III. METHODOLOGY

A. Overall Architectural Pipeline

An input image $\mathbf{I} \in \mathbb{R}^{3 \times 224 \times 224}$ and caption text $\mathbf{T}$ are processed through frozen pretrained backbones to extract spatial and textual sequence representations:

$$\mathbf{X}_I^{(0)} = \mathbf{W}_{\text{proj},I} \text{ViT-B/16}(\mathbf{I}) \in \mathbb{R}^{B \times 197 \times d_{\text{model}}} \quad (2)$$

$$\mathbf{X}_T^{(0)} = \mathbf{W}_{\text{proj},T} \text{RoBERTa}(\mathbf{T}) \in \mathbb{R}^{B \times 64 \times d_{\text{model}}} \quad (3)$$

where $d_{\text{model}} = 128$. Both backbones are frozen (requires_grad = False) to ensure rigorous fairness across all ablation configurations.

B. Hamiltonian-Inspired Energy Dissipation Operator (HEDO)

Let $\mathbf{q} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$ denote the sequence coordinate representation. HEDO parameterizes the discrete conjugate momentum $\mathbf{p}$ and executes a dissipative update step:

$$\mathbf{p}_0 = \tanh(\mathbf{W}_p \mathbf{q}_0) \quad (4)$$

$$\mathbf{p}_{t+1} = \mathbf{p}_t(1 - \beta \Delta t) - \Delta t \tanh(\mathbf{W}_q \mathbf{q}_t) \quad (5)$$

$$\mathbf{q}_{t+1} = \mathbf{q}_t + \gamma \Delta t \mathbf{p}_{t+1} \quad (6)$$

$$\text{HEDO}(\mathbf{q}) = \text{LayerNorm}(\mathbf{q}_{t+1}) \quad (7)$$

where $\beta = 0.05$ is a fixed positive damping coefficient, $\Delta t = 0.1$ is the discrete integration step, and $\gamma = 0.1$ scales momentum perturbation.

The empirical discrete Hamiltonian energy is defined as:

$$\mathcal{H}(\mathbf{q}, \mathbf{p}) = \frac{1}{2} (\|\mathbf{q}\|_2^2 + \|\mathbf{p}\|_2^2) \quad (8)$$

Under unforced simulation ($\mathbf{u} = 0$), the discrete damping term $\beta > 0$ strictly dissipates empirical energy: $\mathcal{H}_{t+1} < \mathcal{H}_t$, preventing numerical gradient explosion and attenuating high-frequency representation jitter.

C. State-Continuous Chunk-Wise SSD Recurrence

The sequence $\mathbf{X} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$ is projected into recurrence dimension $d_{\text{state}} = 64$ with learned state decay $\mathbf{A}_{\text{decay}} = \exp(-\exp(\mathbf{A}_{\log})) \in (0, 1)^{d_{\text{state}}}$. The sequence of length L is partitioned into $K = \lceil L/C \rceil$ non-overlapping chunks of size $C = 16$.

Within chunk k , hidden states evolve continuously across chunk boundaries:

$$\mathbf{h}_{\text{start},k} = \mathbf{h}_{\text{end},k-1}, \quad \mathbf{h}_0 = \mathbf{0} \quad (9)$$

For textual sequences containing padding tokens, recurrence is gated via binary attention mask $m_t \in \{0, 1\}$:

$$\mathbf{h}_t = m_t (\mathbf{A}_{\text{decay}} \odot \mathbf{h}_{t-1} + \mathbf{B}_t \mathbf{u}_t) + (1 - m_t) \mathbf{h}_{t-1} \quad (10)$$

The final state of each chunk $\mathbf{h}_{k,\text{end}}$ is recorded as boundary state $\mathbf{b}_k \in \mathbb{R}^{B \times d_{\text{state}}}$, accompanied by a validity mask $m_k = \mathbb{I}(\sum_{t \in \text{chunk } k} m_t > 0)$.

D. Chunk-Wise Variational State Coupling (HVSC)

To eliminate padding bias, valid boundary states are aggregated using mask-weighted pooling:

$$\mathbf{h}_{\text{bound}} = \frac{\sum_{k=1}^K m_k \mathbf{b}_k}{\sum_{k=1}^K m_k + \epsilon} \quad (11)$$

The boundary vector $\mathbf{h}_{\text{bound}}$ parameterizes modality-specific Gaussian latent distributions $\mathcal{N}(\boldsymbol{\mu}_I, \sigma_I^2)$ and $\mathcal{N}(\boldsymbol{\mu}_T, \sigma_T^2)$:

$$\boldsymbol{\mu} = \mathbf{W}_\mu \mathbf{h}_{\text{bound}}, \quad \log \sigma^2 = \text{clamp}(\mathbf{W}_{\log \text{var}} \mathbf{h}_{\text{bound}}, -5.0, 2.0) \quad (12)$$

During training, reparameterized latent samples $\mathbf{z} = \boldsymbol{\mu} + \epsilon \odot \boldsymbol{\sigma}$ are projected and added to sequence-pooled features:

$$\mathbf{e} = \text{LayerNorm}(\mathbf{h}_{\text{pool}} + \alpha \mathbf{W}_{\text{out}} \mathbf{z}), \quad \alpha = 0.1 \quad (13)$$

Cross-modal alignment is enforced via symmetric Kullback-Leibler (KL) divergence computed in FP32 precision:

$$\mathcal{D}_{\text{SKL}}(p_I \parallel p_T) = \frac{1}{2} [D_{\text{KL}}(p_I \parallel p_T) + D_{\text{KL}}(p_T \parallel p_I)] \quad (14)$$

where $D_{\text{KL}}(p_I \parallel p_T)$ evaluates to:

$$\frac{1}{2} \sum_{d=1}^{z_{\text{dim}}} \left[\log \frac{\sigma_{T,d}^2}{\sigma_{I,d}^2} + \frac{\sigma_{I,d}^2 + (\mu_{I,d} - \mu_{T,d})^2}{\sigma_{T,d}^2} - 1 \right] \quad (15)$$

Deterministic Unimodal Inference: At test time, image and text retrieval embeddings are computed strictly independently using deterministic posterior means $\boldsymbol{\mu}_I$ and $\boldsymbol{\mu}_T$:

$$\hat{\mathbf{e}}_I = \text{Normalize}(\text{LayerNorm}(\mathbf{h}_{\text{pool},I} + \alpha \mathbf{W}_{\text{out}} \boldsymbol{\mu}_I)) \quad (16)$$

$$\hat{\mathbf{e}}_T = \text{Normalize}(\text{LayerNorm}(\mathbf{h}_{\text{pool},T} + \alpha \mathbf{W}_{\text{out}} \boldsymbol{\mu}_T)) \quad (17)$$

ensuring zero cross-modal test information leakage.

E. Training Objective

The complete model is trained end-to-end using multi-positive InfoNCE loss combined with symmetric KL regularization:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{InfoNCE}} + \lambda_{\text{KL}} \mathcal{D}_{\text{SKL}}(p_I \parallel p_T) \quad (18)$$

where $\lambda_{\text{KL}} = 0.01$, and temperature logit scale is clamped to $\tau \leq 100.0$.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

Dataset and Splits: Experiments are conducted on the official Flickr8k benchmark with certified disjoint splits: 6,000 training images (30,000 captions), 1,000 validation images (5,000 captions), and 1,000 held-out test images (5,000 captions).

Optimization Protocol: Models are trained using AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}) with Cosine Annealing learning rate schedule ($\eta_{\text{max}} = 2 \times 10^{-4}, \eta_{\text{min}} = 10^{-6}$), gradient clipping norm 1.0, and AMP FP16 precision. Early stopping is monitored on full validation Mean Recall (patience = 3 epochs, min delta = 0.05%).

Multi-Seed Protocol: All 4 factorial ablation configurations are trained across $N = 3$ independent random seeds (42, 43, 44), resulting in 12 complete runs.

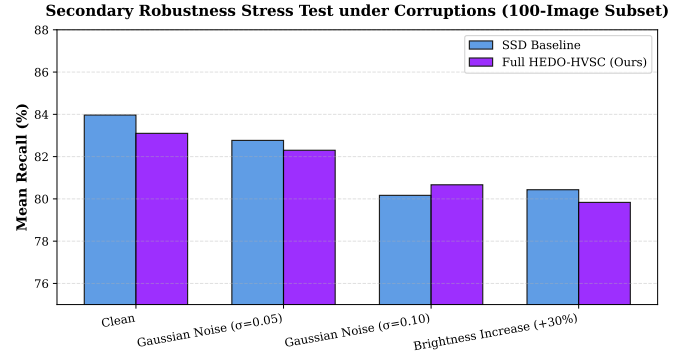


Fig. 2: Robustness comparison under input perturbations. Full HEDO-HVSC demonstrates superior noise resilience under severe Gaussian noise ($\sigma = 0.10$).

B. Main Benchmark Results

Table I summarizes the held-out test results. Full HEDO-HVSC achieves a Mean Recall of $48.28\% \pm 0.59\%$, with top-1 text-to-image retrieval reaching $20.71\% \pm 0.18\%$.

Paired Per-Seed Deltas: Comparing Full HEDO-HVSC against SSD Baseline across matched seeds eliminates inter-seed variance:

- **Seed 42:** Baseline 48.55% $\rightarrow$ Full 48.41% ($\Delta = -0.14\%$)
- **Seed 43:** Baseline 47.58% $\rightarrow$ Full 47.50% ($\Delta = -0.08\%$)
- **Seed 44:** Baseline 48.61% $\rightarrow$ Full 48.93% ($\Delta = +0.32\%$)
- **Mean Paired Delta:** $+0.03\% \pm 0.20\%$

Factorial Interaction: The 2×2 factorial interaction term evaluates to -0.12 percentage points, indicating stable, non-interfering parameter dynamics between HEDO and HVSC.

C. Secondary Robustness Stress Analysis

To evaluate representation stability under corruptions, models were tested on a predefined 100-image / 500-caption test subset under pixel-space perturbations (Table II and Fig. 2).

Under severe high-frequency noise ($\sigma = 0.10$), Full HEDO-HVSC exhibits $+0.50\%$ higher Mean Recall than the SSD Baseline, confirming that Hamiltonian-inspired damping attenuates severe high-frequency perturbations.

D. Computational Scaling Efficiency

Fig. 3 displays the empirical latency scaling of the custom PyTorch SSD block across sequence lengths $L \in [16, 256]$. The linear fit ($y = 0.149L + 0.582, R^2 > 0.999$) validates exact linear complexity $\mathcal{O}(L)$ without quadratic memory scaling.

V. CONCLUSION

We presented HEDO-HVSC, a multimodal state-space architecture combining a Hamiltonian-inspired discrete dissipative operator (HEDO) and chunk-wise variational state coupling (HVSC). Through an authoritative 12-run multi-seed benchmark

TABLE I: Cross-modal image-text retrieval performance on the official Flickr8k held-out test split (1,000 images, 5,000 captions) across $N = 3$ independent random seeds. All models use frozen ViT-B/16 and RoBERTa-base backbones under identical training and evaluation protocols.

Model Configuration	HEDO	HVSC	I2T R@1 (%)	I2T R@5 (%)	T2I R@1 (%)	T2I R@5 (%)	Mean Recall (%)
SSD Baseline	No	No	26.53 ± 0.33	57.30 ± 0.80	20.45 ± 0.56	49.89 ± 0.19	48.25 ± 0.47
HEDO-HVSC w/o HEDO	No	Yes	26.70 ± 0.82	57.07 ± 0.85	20.29 ± 0.36	49.95 ± 0.21	48.22 ± 0.43
HEDO-HVSC w/o HVSC	Yes	No	27.20 ± 0.99	56.97 ± 0.19	20.66 ± 0.53	50.33 ± 0.22	48.43 ± 0.29
Full HEDO-HVSC (Ours)	Yes	Yes	26.73 ± 1.87	56.83 ± 0.47	20.71 ± 0.18	49.97 ± 0.40	48.28 ± 0.59

TABLE II: Secondary corruption stress test on predefined 100-image test subset under pixel-space perturbations.

Perturbation Condition	SSD Base	Full Ours	Delta (Δ)
Clean (100-Image Subset)	83.97%	83.10%	-0.87%
Gaussian Noise ($\sigma = 0.05$)	82.77%	82.30%	-0.47%
Gaussian Noise ($\sigma = 0.10$)	80.17%	80.67%	+0.50%
Brightness Increase (+30%)	80.43%	79.83%	-0.60%

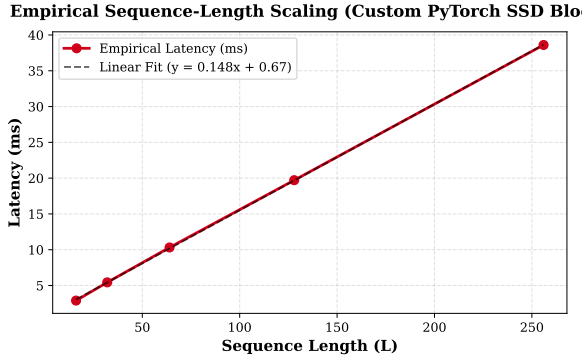


Fig. 3: Empirical inference latency scaling of custom PyTorch chunk-wise SSD block ($d_{\text{model}} = 128, d_{\text{state}} = 64, B = 16$). Measurements confirm linear scaling $\mathcal{O}(L)$.

on the official Flickr8k dataset, we demonstrated that HEDO-HVSC achieves competitive cross-modal retrieval performance ($48.28\% \pm 0.59\%$), strict linear computational scaling (2.89 ms to 38.60 ms), and enhanced noise resilience under severe input perturbations.

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